GenAI-Augmented Asset Prioritization & Automation Workflow

Addendum to: Hybrid Cloud–Blockchain Inventory Management System

Prepared for internal engineering & procurement review — July 2026
Builds on Phase 1 Go prototype and Phase 2 Hyperledger Fabric v2.5 integration

This addendum extends the existing Hybrid Cloud–Blockchain Inventory Management System (Phase 1 Go prototype + Phase 2 Hyperledger Fabric v2.5 network) with three additions requested for enterprise readiness: (1) a formal **asset priority classification framework** (High / Medium / Low, plus the standard procedures that go with it), (2) a set of **Generative-AI automation workflows** layered on top of the existing Demand Forecasting, Anomaly Detection, NL Query and Procurement services, and (3) the concrete **changes required** to the chaincode data model, REST API contracts, RBAC, and the world-state indexes to support all of the above. It is written as a delta document — it assumes the architecture already documented in the base System Overview and the Phase 1/Phase 2 README.

2 Asset Priority Classification Framework

[illegible] frequency, insurance, and AI monitoring intensity scale with how much the business actually depends on the asset — rather than treating every SKU the same. The framework below is criticality-weighted (a standard ABC / criticality-analysis approach used in ISO 55000 asset-management programmes) and is designed to be computed automatically by the new Classification Agent described in Section 4.

2.1 Classification Criteria & Weights

Criterion	Weight	What it measures
[illegible]	30%	Impact on department operations if the asset is unavailable or fails
Replacement Cost	20%	Monetary value / procurement cost of the asset
[illegible]	20%	Time required to source, approve and deploy a replacement
Safety / Compliance Impact	15%	Regulatory exposure, safety-critical function, audit obligation
[illegible]	15%	Whether spares, backups, or alternate assets already exist

Each criterion is scored 1–5 (5 = highest risk/impact) by the classification agent using asset metadata, department, historical consumption pattern and, where available, maintenance history. The weighted sum produces a 1.0–5.0 **Criticality Score** that maps to a tier:

Score Range	Tier	Audit Cadence	AI Monitoring
≥ 4.0	**P1 — HIGH / CRITICAL**	Monthly physical + continuous	Real-time anomaly + predictive maintenance
2.5 – 3.9	**P2 — MEDIUM**	Semi-annual physical check	Batch anomaly scan (weekly)
< 2.5	**P3 — LOW**	Annual / spot-check	None (ledger record only)

2.2 Illustrative Tier Assignment by Department

Tier	Lab	Admin / Office	Central Store	IT / Electronics
[illegible]	Core diagnostic / production machines	Primary domain controller, access-control system	Cold-chain / secure storage units	Core servers, network switches, firewalls
P2 Medium	Secondary lab instruments, calibration tools	Staff workstations, printers	Forklifts, handling equipment, scanners	Peripheral devices, workstations

Tier	Lab	Admin / Office	Central Store	IT / Electronics
P3 Low	Consumables, glassware, PPE	Stationery, furniture	Packing material, small hand tools	Cables, spare parts, accessories

Note: tier assignment is a computed default, not a hard rule — department heads or the Store Manager can override a tier via the new `updatePriorityTier()` *chaincode function, with the justification permanently recorded on-chain for audit purposes.*

3 Standard Asset Management Procedures

The workflow already documents the blockchain-specific transaction lifecycle (issue → consume → transfer → audit). What is missing for a production-grade Enterprise Asset Management (EAM) deployment is the surrounding governance procedure that most ITAM programmes follow (aligned to ISO 55000 asset management, ISO/IEC 19770 IT asset management, and ITIL Asset & Configuration Management practice). The table below is the full lifecycle this system should now enforce, tier-aware.

3.1 Full Asset Lifecycle

Stage	Activity	Owner	On-chain / Off-chain
1. Plan	Demand forecast triggers a procurement request	Demand Svc / Store Mgr	Off-chain (PostgreSQL)
2. Acquire	PO approved, vendor selected, goods received	Procurement Agent / Store Mgr	Off-chain → anchored
3. Tag & Register	QR/RFID tag issued; issueAsset() TX with priorityTier	Dept Staff	On-chain (ledger + world state)
4. Deploy	Asset assigned to department / user / location	Dept Staff	On-chain
5. Maintain	Preventive/corrective maintenance, AMC service visits	IT Admin / Vendor	On-chain event + off-chain ticket
6. Monitor / Utilize	Usage tracked via consumeStock/transferAsset TX stream	System (automatic)	On-chain
7. Depreciate	Book value recalculated on a fixed schedule	Finance / PostgreSQL job	Off-chain
8. Audit	Physical verification against ledger, tier-based cadence	Asset Auditor (new role)	On-chain auditLog + AI vision report
9. Retire / Dispose	Write-off approval, e-waste compliant disposal	Dept Head + IT Admin sign-off	On-chain (dual-signature TX)

3.2 Tagging, Depreciation & Coverage Standards

Tier	Tag Type	Depreciation (useful life)	Insurance / AMC
P1 High	RFID + QR, dual-redundant	5 years, straight-line	Mandatory AMC + insured
P2 Medium	QR code	3 years, straight-line	Recommended AMC
P3 Low	QR / barcode or batch-tagged	1 year or expensed at issue	Not required

Disposal must follow India's E-Waste (Management) Rules, 2022 for IT/electronics assets, and requires a dual-signature retirement transaction (department head + IT Admin) before the chaincode marks an asset record as `RETIRED`. This is a new required state — see Section 6.

4 GenAI Workflow Additions

The base design already includes four Gen-AI microservices (Demand Forecasting, Anomaly Detection, NL Query, Procurement Agent). To fully automate the priority-tiered lifecycle above, nine additional agents/services are proposed. Each is designed to plug into the existing REST API Gateway and Kafka/Pulsar event stream — no change to the Fabric consensus layer itself is required.

4.1 Priority Classification Agent

Function	Auto-assigns / re-evaluates priorityTier from asset metadata + usage history
Trigger / Input	issueAsset TX, periodic re-scan
Model Type	LLM classifier + weighted scoring
Output / Integration	Writes priorityTier via classifyPriority() chaincode fn

4.2 Predictive Maintenance Agent

Function	Forecasts failure risk / next service date from TX + telemetry stream
Trigger / Input	Kafka block-event stream
Model Type	Time-series model (LSTM/Prophet) + LLM summarizer
Output / Integration	Auto-creates maintenance work order, alerts IT Admin

4.3 Vision-Based Audit Verification Agent

Function	Matches a photo of the physical asset/tag against the ledger record during audits
Trigger / Input	Auditor photo upload
Model Type	Vision-capable LLM
Output / Integration	Confirms match / flags discrepancy in auditLog

4.4 Document Intelligence Agent

Function	Extracts structured fields from invoices, warranty cards, AMC contracts
Trigger / Input	File upload to MinIO/S3
Model Type	Vision-LLM / OCR pipeline
Output / Integration	Auto-fills asset fields; SHA-256 hash anchored on-chain

4.5 Dynamic Reorder (EOQ) Optimization

Function	Replaces the static consumeStock() threshold with a demand-aware reorder point
Trigger / Input	Consumption TX stream
Model Type	Prophet/LSTM forecast + LLM reasoning layer
Output / Integration	Updates per-item threshold used by requestReplenishment()

4.6 Utilization & Idle-Asset Detection Agent

Function	Flags underused or idle P1/P2 assets for reallocation or disposal review
Trigger / Input	Scheduled batch over TX history
Model Type	Clustering / anomaly-on-usage model
Output / Integration	Recommendation report to Store Manager dashboard

4.7 Compliance & Audit Report Generator

Function	Drafts periodic compliance reports (DPDP Act, IT Act 43A/72A, e-waste)
Trigger / Input	Scheduled (monthly/quarterly)
Model Type	LLM summarization over PostgreSQL audit tables
Output / Integration	PDF/report pushed to dashboard + compliance archive

4.8 Conversational Asset Assistant

Function	Extends the existing NL Query engine with tool-calling so staff can transact by chat (“issue 5 gloves to Lab”)
Trigger / Input	Chat message
Model Type	LLM agent with function-calling to REST endpoints
Output / Integration	Executes TX after user confirmation step

4.9 Vendor Negotiation Enrichment

Function	Adds negotiation-draft and price-benchmark generation to the existing Procurement Agent
Trigger / Input	Draft PO created
Model Type	LLM + vendor RAG (existing Vector DB)
Output / Integration	Vendor outreach email draft + benchmark note in dashboard

5 Architecture Changes

A new **GenAI Automation Layer** is inserted between the existing Cloud/AI layer and Cloud DB layer. It consumes the same Kafka/Pulsar block-event stream already produced by the Fabric Gateway + Event Hub, so no change is required to the ordering service, peers, or chaincode endorsement policy.

New Component	Sits Between	Notes
Classification Service	REST API Gateway ↔ Chaincode	Synchronous call on issueAsset; async re-scan job nightly
Predictive Maintenance Consumer	Kafka stream ↔ PostgreSQL	New consumer group; writes to maintenance_predictions table
Vision/OCR Service	File upload API ↔ MinIO/S3	Vision-LLM calls; hash still anchored on-chain as today
Utilization Batch Job	CouchDB world state ↔ Dashboard	Scheduled (daily); read-only, no chaincode change
Compliance Report Job	PostgreSQL ↔ Dashboard/Archive	Scheduled; reads audit + anomaly tables

Optional, tier-gated: P1 assets with existing IoT sensors (e.g. lab equipment with usage counters) can feed a lightweight telemetry topic into the Predictive Maintenance Consumer. This is additive and is not required for P2/P3 assets.

6 Chaincode & Data Model Changes

6.1 New Fields on the Asset Record (chaincode/assetcc)

Field	Type	Set By
priorityTier	enum: P1 | P2 | P3	classifyPriority() / updatePriorityTier()
criticalityScore	float (1.0–5.0)	classifyPriority()
lastAuditDate	RFC3339 timestamp	recordAuditResult()
utilizationRate	float (0–1)	Utilization batch job (off-chain → hash anchored)
warrantyExpiry / amcExpiry	RFC3339 date	Document Intelligence Agent (on issueAsset / update)
lifecycleState	enum: ACTIVE | MAINTENANCE | IDLE | RETIRED	recordAuditResult() / retireAsset()

6.2 New Chaincode Functions

Function	Purpose	Endorsement
classifyPriority(assetID)	Sets initial priorityTier + criticalityScore at issuance	2-of-4, same as issueAsset
updatePriorityTier(assetID, tier, justification)	Manual override of tier by an authorised auditor/manager	2-of-4 + auditor signature
scheduleAudit(assetID, dueDate)	Sets the next mandatory audit date per tier cadence	1-of-4 (informational)
recordAuditResult(assetID, status, evidenceHash)	Logs physical/vision audit outcome, updates lastAuditDate	2-of-4

Function	Purpose	Endorsement
retireAsset(assetID, deptHeadSig, itAdminSig)	Dual-signature write-off / disposal transaction	2-of-4, both signatures required

6.3 CouchDB & Off-chain Schema Changes

- New composite CouchDB index on (priorityTier, deptID) to support tier-scoped dashboard queries.
- New PostgreSQL tables: `asset_utilization`, `maintenance_predictions`, `audit_log_ai`, `vendor_negotiation_log`.
- Vector DB namespace extended to store embeddings for warranty/AMC document search (Document Intelligence Agent).

7 Updated API Contracts

New endpoints to add to the REST API Gateway, all secured via JWT + RBAC unless noted.

Endpoint	Method	Auth	Request Body	Response
/api/assets/classify	POST	JWT+RBAC	{assetID}	priorityTier, criticalityScore
/api/assets/{id}/priority	PATCH	JWT+RBAC (auditor)	{tier, justification}	txID, updatedTier
/api/maintenance/predict	GET	JWT	—	riskScore, predictedServiceDate
/api/audit/vision-verify	POST	JWT+RBAC (auditor)	multipart: photo + assetID	matchStatus, discrepancy[]
/api/assets/utilization	GET	JWT	{deptID, tier}	utilizationRate[], idleAssets[]
/api/compliance/report	GET	JWT+RBAC (admin)	{period}	reportURL, summary
/api/assistant/chat	POST	JWT	{message, sessionID}	reply, proposedAction, requiresConfirm

8 RBAC & Role Changes

Role	New / Changed	Access Granted
asset-auditor	New	scheduleAudit, recordAuditResult, /api/audit/vision-verify, updatePriorityTier
ai-ops	New (service account)	classifyPriority, predictive-maintenance writes, utilization batch job
store-manager	Extended	Adds read access to /api/assets/utilization and idle-asset recommendations
dept-user	Extended	Adds /api/assistant/chat for confirmed self-service transactions
admin (IT Admin)	Extended	Adds /api/compliance/report and retireAsset dual-signature capability

9 Indicative Incremental Cost (India)

Rough monthly addition on top of the existing India cloud costing (Section 8–13 of the base document). These are planning-level estimates for a mid-scale deployment (~200 users, 50K TX/month) and should be validated against actual vendor pricing at contracting time.

New GenAI Component	Basis	Approx. Monthly (■)
Vision-LLM audit + OCR calls	~150K tokens/month, vision-capable model	4,500 – 6,500
Predictive maintenance compute	Time-series model, batch + streaming inference	3,000 – 5,000
Additional vector storage / embeddings	Warranty/AMC document corpus	1,200 – 2,000
Extra orchestration / Kafka consumer group	Small additional compute footprint	2,500 – 4,000
Conversational assistant (tool-calling LLM)	~300K tokens/month	6,500 – 9,000
TOTAL (approx.)		**■17,700 – 26,500 / month**

Estimate only; not a quote. Validate against actual managed-LLM and compute pricing at contract time.

10 Summary & Phased Rollout Recommendation

The additions above are intentionally layered so that none of them require changes to Fabric consensus, endorsement policy, or the existing four Gen-AI services — they extend the data model (new asset fields + chaincode functions), the API surface, and add new stateless/batch services that read the same Kafka event stream and CouchDB world state already in production.

Phase	Scope	Rationale
3a — Foundation	Priority classification framework, chaincode field/function additions, dynamic reorder (EOQ)	Low technical risk, immediate value: better audit targeting, less over/under-stocking
3b — Assurance	Predictive maintenance, vision-based audit verification	Requires new consumer services and vision-LLM integration; builds on 3a's tier data
3c — Experience	Conversational asset assistant, compliance report generator, vendor negotiation enrichment	Highest user-facing value; benefits from mature tier/audit data produced by 3a/3b

Recommended immediate next step: implement the ***priorityTier*** *field and* `classifyPriority()` *chaincode function first (Phase 3a). Every other addition in this document — audit cadence, dynamic reorder, predictive maintenance targeting, and utilization reporting — depends on assets already carrying a tier.*

Document prepared for internal engineering review. Figures and timelines are planning-level estimates; validate against actual vendor SLAs and pricing before committing to a rollout schedule.